

Research paper

High-repetition-frequency high-power pulsed laser diode driver

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ABSTRACT

A wide-range high-speed high-resolution LiDAR requires a laser source with high power and high repetition frequency, where the laser power driver plays an important role. Here we propose a high-repetition-frequency high-power pulse power supply for laser diode driving by using frequency synthesis technique. This technique generates a high repetition frequency pulse through synthesizing the driving signals for multiple parallel switching transistors. The output characteristics of this pulse power supply were evaluated under different loads including a resistor, a single laser diode, a laser diode array. The results demonstrate that the proposed circuit is capable of outputting a pulse current of 10 A and a repetition frequency of 10 MHz at different loads, with a rise time of 2.25 ns. Finally, the output characteristics of the laser at different repetition frequencies were measured. A maximum peak power of 90 W was achieved. Additionally, a pulsed laser diode with nano-tunnelling technology shows that the driver has the potential for higher repetition frequency.

1. Introduction

Light detection and ranging (LiDAR) technology has widely been used in military [1,2], driverless technology [3], aerospace industry [4–6] and industrial manufacturing [7–9]. Moreover, increasing application scenarios impose higher demands on laser-based LiDAR system, such as faster processing speeds, longer detection ranges, and greater environmental adaptability [9–12]. The laser power supply, also known as laser driver, is an essential part of LiDAR system. The improvement of LiDAR performance has put higher requirements on the driver such as faster repetition rate, higher maximum output power, and wider output range [13–15].

In recent years, many methods have been proposed to generate short power pulses. Some researchers used several kinds of specific semiconductor devices such as metal oxide-semiconductor field-effect transistors (MOSFETs) [16–18], avalanche diodes, and fast recovery diodes [19,20] to generate short pulses. Wang et al. designed a short pulse generator and a short pulse driver to provide pulses with a repetition frequency of 5 kHz to 25 MHz and a pulse width of 3.5 ns to 51 ns, continuously adjustable. However, the output power is only 7.94 mW

and the output peak power cannot be controlled. Cho et al. proposed a pulsed laser driver using an avalanche transistor and switches for LiDAR application. However, the repetition frequency is 50 kHz, and feedback control is not possible [21]. In addition to the single-stage avalanche transistor, Marx generators [22,23] and multilevel converters [24–26] are popular in the field of short pulse generation. For example, Chen et al. proposes a pulse generator with the characteristics of ultra-wide-band, high-amplitude, and bipolar output, which contains a bipolar output circuit combining the Marx circuit and the pulse narrowing circuit. However, the Marx generator is mostly used to generate high voltage pulses and is rarely used in current devices. It is not suitable for high-repetition-frequency circuits, especially those operating above megahertz.

Besides, the heat dissipation due to switching loss puts another limitation for high-repetition operation in laser drivers. With the increase of switching frequency, the switching loss also increases gradually. Traditional switching devices, such as power MOSFETs or insulated gate bipolar transistors (IGBT), have high switching loss when operating at high frequency, which leads to a decrease in the output efficiency of the power supply or even a fatal damage to the power device [27]. Liu

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et al. employed the depletion-mode gallium nitride high-electron-mobility transistor (D-mode GaN HEMT) to generate nanosecond pulses at a repetition rate up to 10 MHz for a vertical-cavity surface-emitting laser (VCSEL) diode, of which the maximum peak power reached 150 W [28]. Sánchez et al. achieved a much higher peak power of 880 W with 6 ns pulse width through applying excessive high current with no damage occurred to the laser diode [29]. However, the performance of the driver was only evaluated with burst pulses, and its capability of continuous high-frequency high-power driving remains unknown.

In this article, a pulse power supply with high repetition frequency and high power was designed. The system consists of a pulse power module and a direct current (DC) power supply with programmable output current and voltage for the pulse module. This system can achieve higher output power, repetition frequency and wider adjustable pulse width while using commercial GaN driver transistors, without affecting the optical output of the laser. Multi-channel pulse synthesis technique and enhancement-mode GaN transistors are employed in the system to reduce heat accumulation and the performance requirements of a single transistor, while ensuring high-repetition-frequency fast-rise-time short pulse generation. Finally, an evaluation system with multiple loads was built to evaluate the pulse power supply. Experimental results show that the pulse laser supply can provide pulses with a repetition frequency of 10 kHz to 10 MHz and a pulse width of 10 ns to 1 μ s. The maximum repetition frequency can reach 40 MHz when the load is a nano-tunnelling laser.

2. Design

The power supply is specifically designed for a laser of which the specifications require a maximum current of 10 A, a repetition frequency up to 10 MHz, and a duty cycle ranging from 0 % to 50 %. This design meets the higher demands on wide range of pulse width, high power, and high repetition frequency. However, traditional driving circuit cannot achieve all the specifications above simultaneously, due to the speed and power-dissipation limitations of switching transistor. A higher repetition frequency results in higher switching losses in the transistors. In applications with high power and high repetition frequency, a single power transistor may fail to operate properly due to the accumulation of heat. The optimized multiple-parallel-switch circuit is perfectly suitable for the design goal.

The pulse driver system consists of a DC power supply, a pulse generator, and a field programmable gate array (FPGA) as the main controller, as shown in Fig. 1.

The DC power supply incorporates a buck-boost controller for power regulation, with external programmability. The output of DC power supply is connected to a capacitor array to reduce the output ripple of the DC power circuit, and to achieve a better load response since the high load current is pulsed.

The short pulse generator is composed of multiple parallel switching transistors and a current detection circuit. The current detection circuit samples the pulse current signal and sends it to the control circuit, which regulates the output of the pulse power module. The FPGA is selected as the system control chip. FPGA regulates the output current and voltage of the DC circuit by controlling the multi-channel DAC output. It also generates the driving signal of the switching transistor.

2.1. DC power supply

In the DC power supply module, the control chip of the DC-DC converter is the most crucial. On one hand, the converter is required to ensure the maximum output power of the laser. On the other hand, since it is necessary for the FPGA to control the output voltage and current of the DC power supply, the LT8390 chip with SS (soft start) pin and CTRL pin is selected.

LT8390 is a synchronous 4-switch buck-boost DC-DC controller integrated circuit (IC). The voltage and average current output are adjustable by applying controlling voltages on the SS pin and CTRL pin of the IC, respectively. The CTRL pin is used to program the output current limit of the IC, as shown in Fig. 2(a). We connect this pin to one output channel of a 12-bit quad-channel digital-to-analog converter (DAC, MS4224D) for output current adjustment. The SS pin is originally designed to function as a soft-start control pin, of which the circuit diagram inside the IC is shown in Fig. 2(b). If a capacitor is connected to

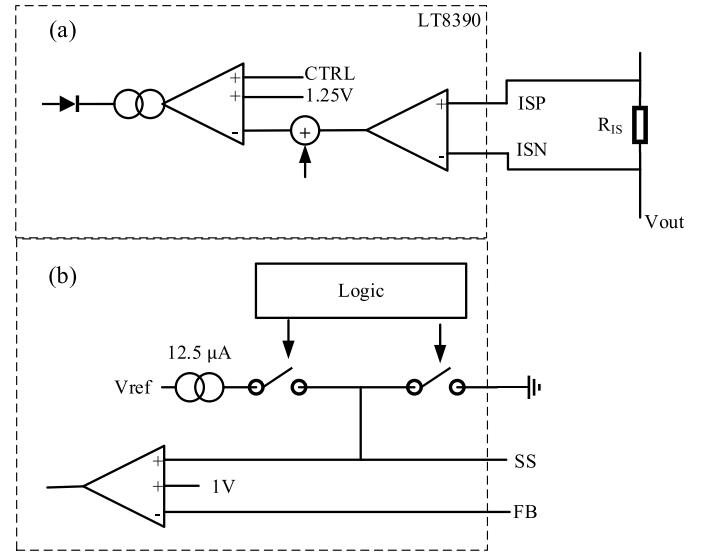


Fig. 2. LT8390 Circuit diagram. (a) CTRL pin function diagram. (b) SS pin function diagram.

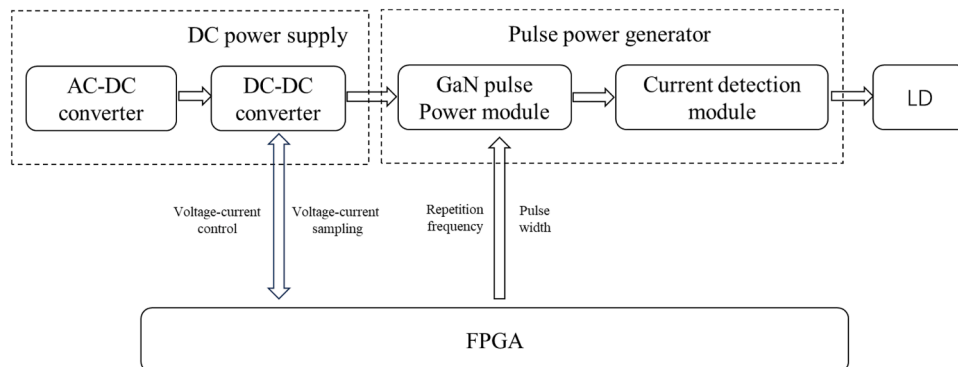


Fig. 1. Pulse driver supply system block diagram. FPGA: field-programmable gate array.

this pin as per its datasheet, the voltage at the pin as well as the output ramp up gradually; and when the voltage at SS pin exceeds 1.75 V, the IC runs normally. In this design, we connect the SS pin to the output of another channel of the DAC, of which the output ranges from 0 V to 1 V. As a result, LT8390 will never enter the normal operation state but is under control of the voltage at SS pin. Thus, we can control both the voltage and current output of the DC circuit by adjusting the outputs of the DAC.

2.2. Short pulse generator

The diagram of the proposed short pulse generator circuit is shown in Fig. 3. S_1 to S_m are power switch transistors. R_S is placed between the source of power transistors and the ground, which functions as a current sampling resistor. Signal 1 to Signal m are the control signals of the power transistors. V_{in} is the voltage source from the DC power supply. In order to effectively lower the voltage and discharge the capacitors as required when the LD is held OFF, a discharge path composed of S_L and R_L is built. When the system control starts to turn off, S_L is turned on, the capacitors are quickly discharged, and at the meantime V_{in} decreases to 0.

A frequency synthesis method is proposed to generate high-repetition-frequency short pulses. The signal timing diagram of frequency synthesis control is shown in Fig. 4. A high repetition frequency pulse signal is generated by combining multiple low-repetition-frequency signals with different values of delay time. T_{on1} , T_{on2} , ..., T_{onm} are the ON time of the corresponding power transistors S_1 , S_2 , ..., S_m , and the values are equal. Similarly, T_{L1} , T_{L2} , ..., T_{Lm} are the time delay between adjacent control signal pulses, and they are also equal. The turn-on time of the power pulse output is determined by the turn-on time of the single transistor, and the repetition frequency is m times the repetition frequency of the single transistor, where m is the number of power transistors. T_{cycle} Indicates the cycle of the control signal, and $T_{cycle} = m \times (T_{onm} + T_{Lm})$. Within each cycle, when one switching transistor is on, the others remain off. This parallel driving scheme effectively reduces the switching frequency of each transistor, which reduces heat dissipation of individual transistors.

In the pulse generator module, the most crucial component is the fast-switching transistor device. Due to the requirement of a repetition frequency of 10 MHz for the power supply design, a GaN device with lower switching loss and faster switching speed was selected as the switching transistor. When selecting GaN FETs, first of all, the switching transistor is required to have higher working voltage and current than the maximum voltage and current of the laser, respectively. Secondly, the one with the smallest input and output capacitance is optimum, where the former plays a more important role. In this circuit, a GaN driver (LMG1025) was used as the switching transistor driver, and GaN transistors (EPC2204A) were selected. EPC2204A was selected because it has a remarkably low Q_g , while fulfilling the requirement of the maximum current and voltage of the system, which contributes to a shorter switching time and lower energy consumption under the same driving conditions.

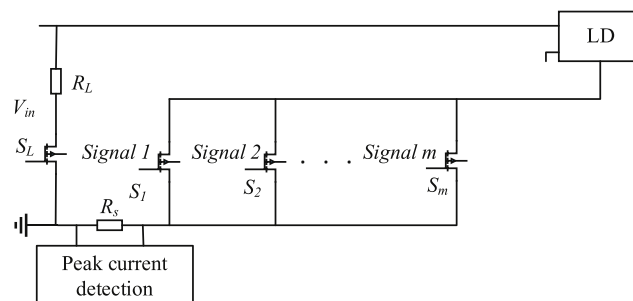


Fig. 3. Pulse drive diagram.

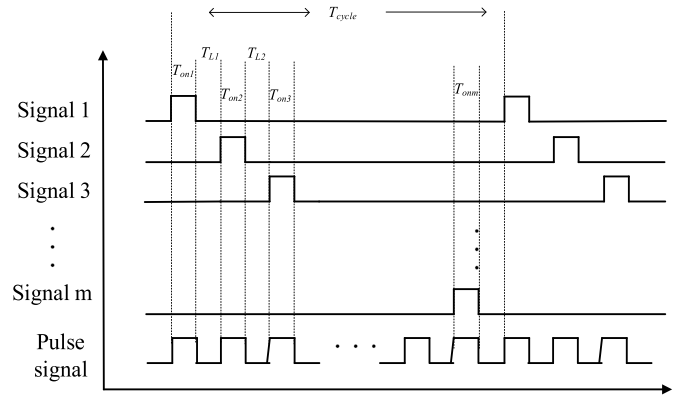


Fig. 4. Control signal timing diagram.

2.3. Analysis of circuit models

The rate equation in the active region of semiconductor laser reflects the relationship between the average carrier density and the average photon density, which can be expressed as

$$\begin{cases} \frac{dN}{dt} = \frac{1}{qV} - \frac{N}{\tau_n} - \frac{g_0(N - N_t)}{1 + \epsilon S} S \\ \frac{dS}{dt} = \Gamma \frac{g_0(N - N_t)}{1 + \epsilon S} S - \frac{S}{\tau_p} + \Gamma \beta \frac{N}{\tau_n} \end{cases} \quad (1)$$

where N is the average carrier density, S is the average photon density, V is the volume of the active region, I is the driving current of the injected laser, q is the charge of the electron, ϵ is the optical nonlinear gain compression factor, β is the spontaneous radiation component, Γ is an optical limiting factor, N_t is the density of the carrier, τ_n is the lifetime of the carrier, τ_p is the lifetime of the photon, g_0 is the rated optical gain.

When the laser reaches a stable state, the corresponding intrinsic circuit model can be analyzed using the steady state of the rate equation. At this moment, the carrier density and photon density in the resonator do not change with time and reach a steady state. Now $\frac{dN}{dt} = 0$, $\frac{dS}{dt} = 0$. The equation can be written as

$$\begin{cases} \frac{1}{qV} = \frac{N}{\tau_n} + \frac{g_0(N - N_t)}{1 + \epsilon S} S \\ \frac{S}{\tau_p} = \Gamma \frac{g_0(N - N_t)}{1 + \epsilon S} S + \Gamma \beta \frac{N}{\tau_n} \end{cases} \quad (2)$$

All the variables need to be decomposed into two forms of DC component and alternating current (AC) component, and then the rate equation is transformed

$$\begin{cases} I = I_0 + i e^{i\omega t} \\ V_j = V_{j0} + v_j e^{i\omega t} \\ N = N_0 + n e^{i\omega t} \\ S = S_0 + s e^{i\omega t} \end{cases} \quad (3)$$

The capital letter on the right side of the equation represents the DC component of the variable, which is the voltage of the PN junction. According to the Shockley equation, the relationship between junction voltage and drive current can be described as (4).

$$I = I_{ss} \exp \left[\left(\frac{qV_j}{2kT} \right) - 1 \right]. \quad (4)$$

Because the product of small signal increments is very small and can be ignored, Eq. (4) can be normalized, and after simplification, the intrinsic equivalent circuit model expression of semiconductor laser can be obtained as follows:

$$\begin{cases} i = i_s + \left(\frac{1}{R_a} + j\omega C_d \right) \nu_j, \\ \nu_j = (R_b + j\omega L) i_s, \end{cases} \quad (5)$$

where $R_a = \frac{R_m}{1 + \tau_n g_0 S_0}$, $R_b = \frac{\epsilon R_m}{\tau_n g_0} + \frac{\beta I R_m \tau_p I_{th}}{q V \tau_n g_0 S_0^2}$, $R_m = \frac{2kT}{q I_{th}}$, $C_d = \frac{\tau_n}{R_m}$, $L = \frac{R_m \tau_p}{\tau_n g_0 S_0}$.

The threshold current I_{th} and the average photon density DC component are

$$\begin{cases} I_{th} = \frac{qV}{\tau_n} \left(\frac{1 + N_i \tau_p \Gamma g_0}{\tau_p \Gamma g_0} \right) \\ S_0 = \frac{\tau_p \Gamma}{qV} (I - I_{th}) \end{cases} \quad (6)$$

Based on the analysis above, the equivalent model of the semiconductor laser diode is shown in Fig. 5, where $R_L = R_a + R_b$.

In the driver circuit, capacitor energy storage is employed to achieve high-power pulse output within a short period of time. The overall circuit model is shown in the Fig. 6.

Here, C_L represents the internal parasitic capacitance of the laser, L_2 represents the internal parasitic inductance of the laser, and R_L represents the equivalent load resistance of the laser. L_1 is the inductance of the circuit line of the circuit board. In this design, a parallel transistor array is employed, which exhibits a higher body capacitance than a single switching transistor. When the switch is turned on, the voltage at point 'a' is equal to the value at point 'b'. At this moment, diode D_1 remains non-conductive, and the laser current I_{SS} stays zero. Conversely, when the switch is turned off, the voltage at point 'a' exceeds that at point 'b', causing diode D_1 to conduct and the laser to generate current. Capacitors C_L and C_{in} jointly supply power to the laser, leading to a situation in which the voltage at point 'b' drops below zero. Consequently, the measured current may exhibit an overcurrent condition due to the discharge of C_L and the distributed inductance L_1 .

2.4. Output current sensing

In the DC power supply, the output voltage and average current are set by the FPGA, and under closed-loop control of the DC-DC controller IC. However, the average current is affected by repetition frequency, voltage applied, and pulse width, and hence the peak pulse current is not directly under control with merely the DC-DC controller.

As is required, a peak current detection circuit is designed, as shown in Fig. 7. A sampling resistor R_1 is connected between ground and source of GaN transistor. The output of the sampling resistor is connected to a high-speed fully-differential amplifier (FDA). R_2 and R_3 are the input

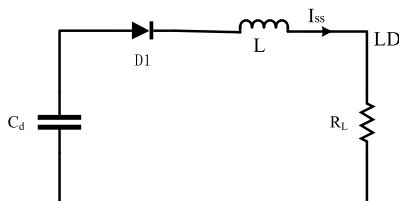


Fig. 5. Model of the semiconductor laser diode.

resistors of the FDA. R_4 and R_5 are the feedback resistors of the FDA. The gain of the FDA is set to 10, where $R_1 = R_2 = 49.9 \Omega$, $R_4 = R_5 = 499 \Omega$. The output of this differential amplifier is then divided into two branches. One branch is fed into a high-speed analog-to-digital converter (ADC) for closed-loop control of the peak current; the other branch feeds the signal into a differential-to-single-ended converting amplifier for measurements with the oscilloscope.

In the peak current detection module, in order to accurately acquire the peak current signal, a fully-differential amplifier LMH5401 was selected. This amplifier has a bandwidth of 8 GHz, and the fully-differential structure provides a better signal-to-noise performance, which meets the design requirements. The maximum repetition frequency of the system is designed to be up to 50 MHz, and hence ADC3564 is selected as the high-speed sampling ADC with a maximum sampling frequency of 125 MSps and 14-bit resolution to meet the design requirements.

2.5. System control

The control module of the power system is designed to control pulse characteristics such as pulse amplitude, pulse width and pulse repetition frequency, as shown in Fig. 8. The trigger source can be generated by internal clock or in response to an external signal.

In addition to control the DC power supply, the FPGA controls the quad-channel DAC (only two channels in use), the dual-channel low-speed ADC, and the high-speed ADC. The low-speed ADC and DAC channels are used to acquire and control the output voltage and average current of the aforementioned DC supply.

The design supports both average current control method and peak current control method. The output of ISMON pin of DC-DC control chip is sampled as the average current of the LD since all the current goes through the load. The peak current signal is acquired by the ADC and is input into the FPGA to be averaged, and is further used for the proportional-integral-differential (PID) control for peak current control. The DACs control signals, which are obtained by PID controller, is realized to control the peak current of the laser driver output.

In this system, the PWM signal achieves a maximum resolution of 1 ns. Theoretically, this requires the FPGA output frequency to reach 1 GHz, while the internal FPGA frequency is limited to 625 MHz, not to mention that the pulse generation logic within the FPGA cannot run under such high clocking frequency. To overcome this limitation, the system employs the SerDes primitive with two synchronized clocks in the FPGA to reach 1 ns timing resolution. A 125 MHz main clock frequency is used, divided from a 500 MHz clock, and 8-bit parallel data is provided to the parallel input port of the SerDes. The output clock of the SerDes is provided as 500 MHz, with double-data-rate (DDR) output capability. The SerDes processes the 125 MHz, 8-bit data in parallel and outputs the data of 1-bit in series with the 500 MHz clock and DDR output, and hence an equivalent output of 1 ns precision is achieved.

2.6. Simulation of driver circuit

The laser diode SPICE model is based on the laser diode model specified in Table 1. The driver circuit of the PSPICE circuit model is shown in Fig. 9.

From the simulation results, it can be observed that when the repetition frequency is 10 MHz, there is a significant voltage fluctuation in the cathode voltage. And such large oscillations will not occur in the output current. This might be caused by the parasitic capacitance and inductance of the laser, resulting in resonance effects. However, such simulations may yield different experimental results due to the inaccuracy of the circuit board and laser parameters.

3. Experimental results and discussion

In order to verify the output characteristics of the proposed driving

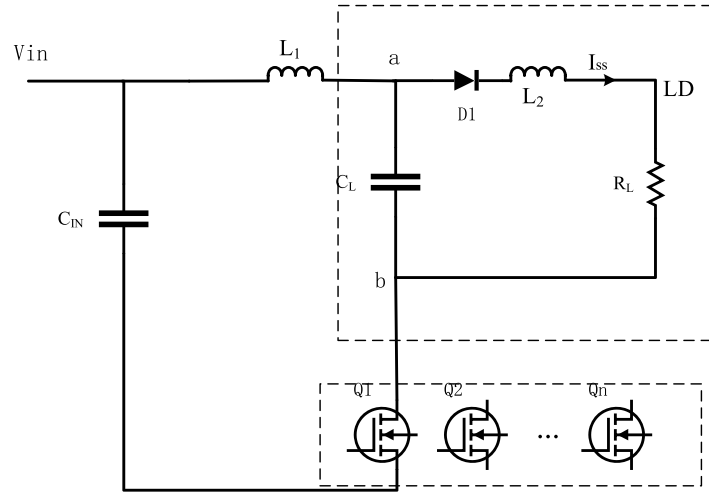


Fig. 6. Driver circuit model.

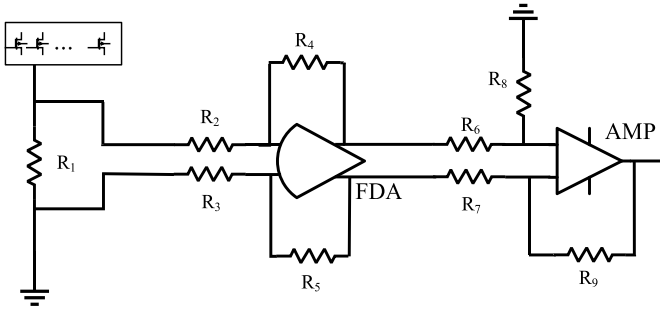


Fig. 7. Peak current detection circuit.

circuit, a measurement system was built, as shown in Fig. 10. The output of DC power supply (Tektronics 2231A-30-3) connected to the DC-DC converter. The output of the DC-DC converter is used to power the pulse power circuit and the FPGA. The supply voltage of the system is set to 12 V. A 1 GHz oscilloscope (LeCroy HDO6104) was selected to sample the waveform.

Three different loads are selected to verify the characteristics of the system. They are 1 Ω resistor (1 W, 1210 package) as a resistive load, a

small but fast single laser diode (PLD-100W-M, Shenzhen Julian Optoelectronics Technology), and a high-power laser diode array (BEJD002, Weifang Academy of Advanced Opto-Electronic Circuits). We added the technical specifications of the array in Supplementary Material.

Due to the power limitation of the fast lower-power load such as the single laser diode and the small resistor, we adopted a pulse-burst strategy for the experiments with such type of load. As shown in Fig. 11. Experimental results of the resistor are demonstrated in Supplementary Material.

The outer envelope of the pulse burst is used to limit the total output power applied to the load. T_1 is the burst width of outer envelope and T_2

Table 1

Parameters for laser diode driver simulation.

Symbol	Unit	Value
L1	nH	20
L2	nH	40
C1	μ F	470
C2	pF	400
R1	R1	1
R2	R2	0.15
T_{width}	ns	50
F_{req}	MHz	10

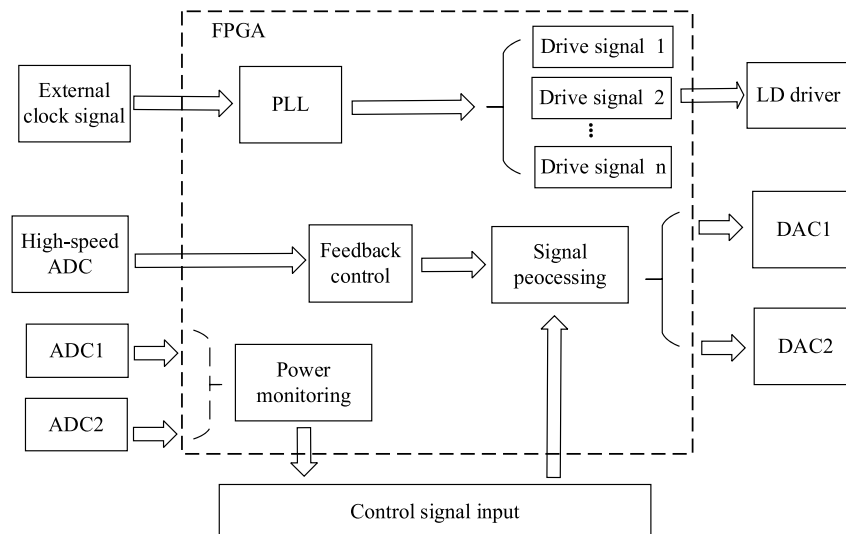


Fig. 8. Control module block diagram.

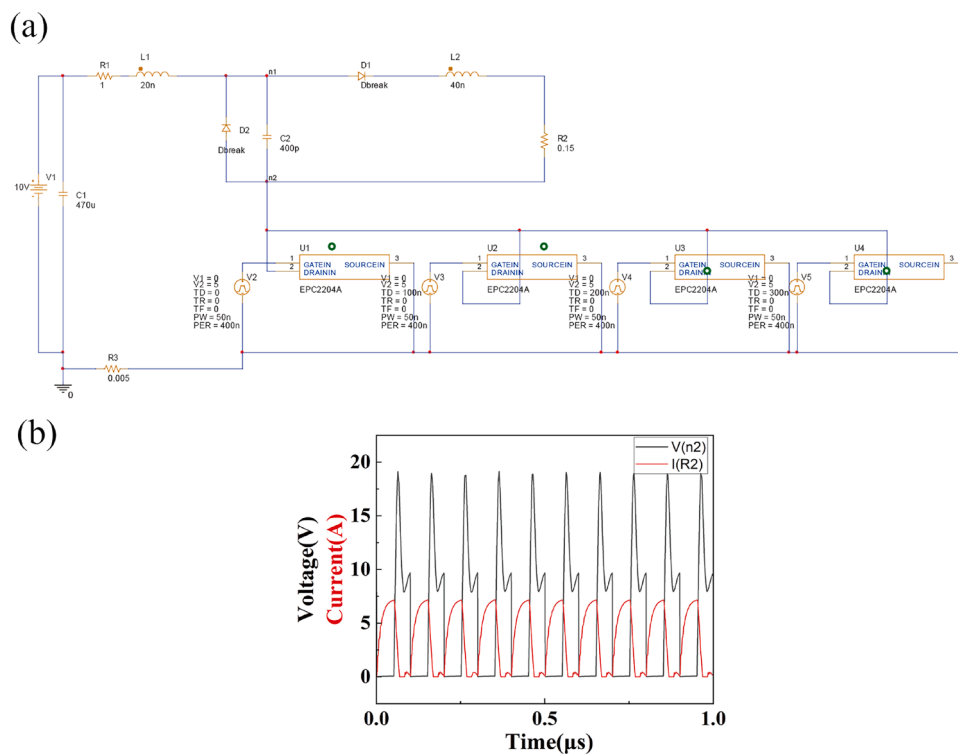


Fig. 9. PSPICE simulation (a) circuit and (b) result for pulse duration 10 ns at 10 MHz.

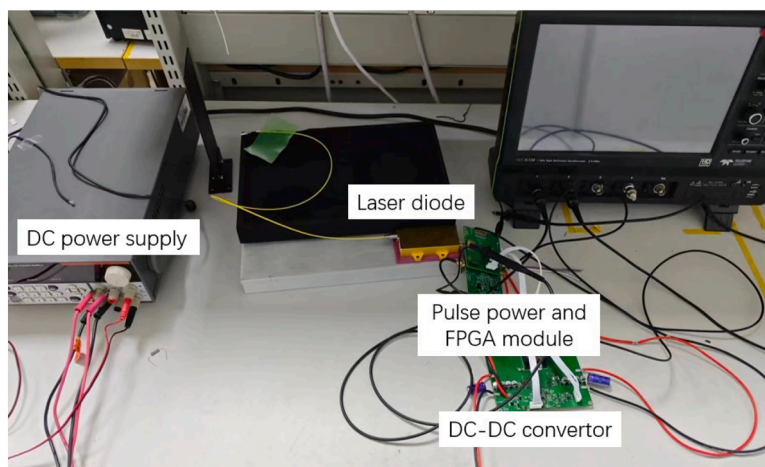


Fig. 10. Photograph of the experimental setup.

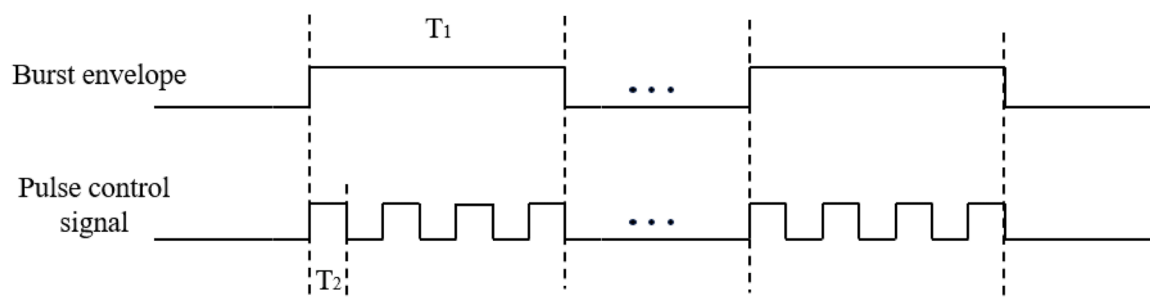


Fig. 11. Power supply control diagram.

is the laser pulse width. In the experiment, the frequency of the outer envelope of the burst is 1 Hz, and the burst width T_1 is 3 μ s. The pulse repetition frequency and pulse width T_2 within the envelope are determined specifically in the experiments as below.

3.1. The voltage and current waveforms of LD at different repetition frequencies

First, the electric characteristics at the load of laser diode array were evaluated with various repetition frequencies. The cathode voltage and current of the laser diode array were measured at the supply voltage of 13 V and different repetition frequencies.

As shown in Fig. 12, at lower repetition frequencies, the measured cathode voltage exceeds the supply voltage. This is due to the inductor in the LD that causes the circuit to behave like a boost converter, leading to a voltage increase. However, as the repetition frequency rises, the internal LC network may fail to fully discharge, resulting in a reduction of the cathode voltage.

Fig. 13 illustrates that the output peak current decreases significantly with increasing repetition frequency. This can be attributed to two factors. First, although the overall activation time at high repetition frequency is the same as that at low repetition frequency, the reduced pulse width leads to a lower peak current. Second, as Figs. 12 and 13 indicate, when the repetition frequency reaches 10 MHz, the voltage and current outputs are not able to reach a stable value before they fall corresponding to the input. This is because, at high repetition frequencies, the self-resonant effects of the parasitic capacitance and inductance inside the laser become increasingly significant, preventing the establishment of a stable current output. As can be seen from Fig. 13, the current waveform shows significant oscillations, but such oscillations were not observed in the subsequent optical tests. By comparing the resistive-load test results in Supplementary Materials, we hypothesize that such oscillations are caused by the driving signal passing through the gate-source capacitance of the power transistor and do not exist in the laser output current.

3.2. The voltage and current waveforms of LD under different voltage

Then, the electric characteristics at the load of laser diode array were evaluated with various driving voltages from 11 V to 14 V, at a fixed repetition frequency of 10 MHz, and the duty cycle is set to 50 %. The voltage at the cathode and the current waveforms are shown in Figs. 14 and 15, respectively. According to the datasheet of the laser, the voltage drop of the laser is approximately 10.5 V. When the laser is turned off, the formula for the cathode voltage of the laser is:

$$V_{cath} = V_{in} - IR_L - V_F \quad (7)$$

where V_{cath} is cathode voltage of laser, V_{in} is input voltage, I is current of laser, R_L is equivalent load resistance of the laser, V_F is forward voltage of laser. As shown in Fig. 14, the cathode voltage within the “off” period is greater than the calculated result. This is due to the parasitic inductance, capacitance inside the laser and flyback diode forming a boost circuit.

As the increase of the supply voltage, the laser diodes within the array were fully switched on, with an apparent rise of the peak current, as shown in Fig. 15. The output current shows obvious oscillation. This is because when the repetition frequency is 10 MHz, the parasitic inductance and capacitance inside the laser generate a resonant effect, resulting in ringing phenomena at the edges of the current pulse.

3.3. Measurement of the influence of the number of parallel switching transistors on the pulse

The fall time of voltage signal at the driver output was measured with 4 and 8 parallel switch transistors, respectively. The supply voltage of the power supply is set to 4 V and the repetition frequency is 10 MHz. The 1 Ω resistor is used as load to minimize the effect of parasitic parameters. As shown in Fig. 16, the fall time of the cathode electrical voltage is approximately 2.25 ns, and it does not change with the increase of the number of parallel switch transistors. Nevertheless, an increase in the number of parallel switch transistors with multiplied output capacitance leads to a larger oscillation. Therefore, the increase in the number of parallel switch transistors does not always lead to an increase in repetition frequency, but with a compromise of stability. In

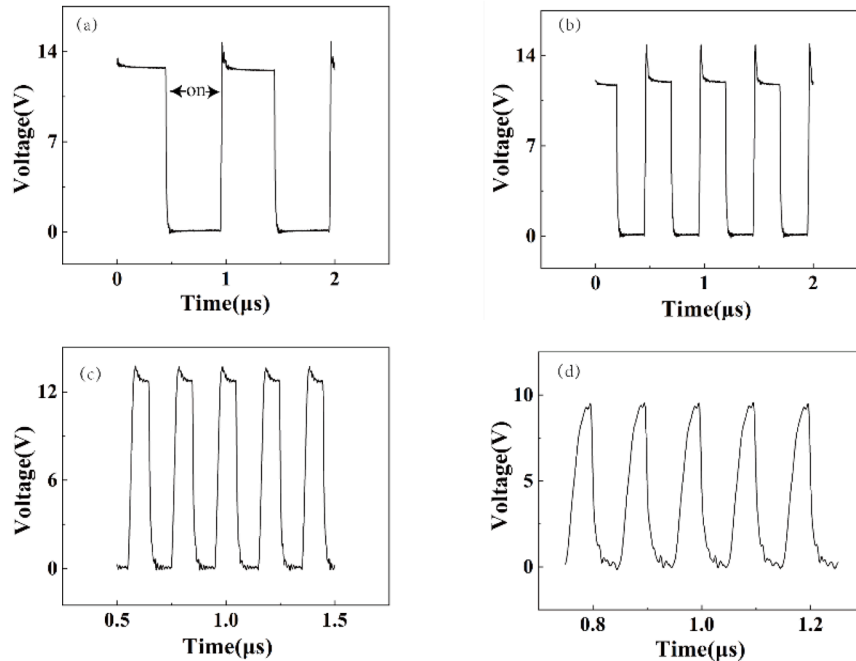


Fig. 12. Cathode Voltage waveform of the laser diode array at different repetition frequencies under 13 V; (a) 1 MHz, (b) 2 MHz (c) 5 MHz, (d) 10 MHz.

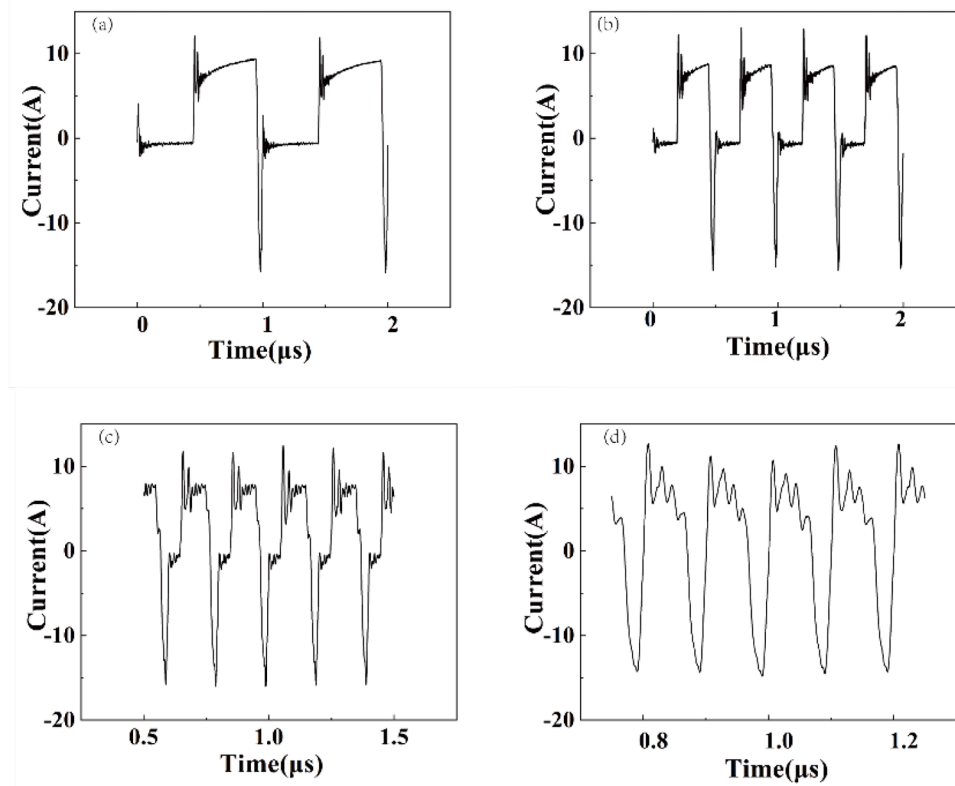


Fig. 13. Current waveform of laser diode array at different repetition frequencies under 13 V; (a)1 MHz, (b)2 MHz (c)5 MHz, (d)10 MHz.

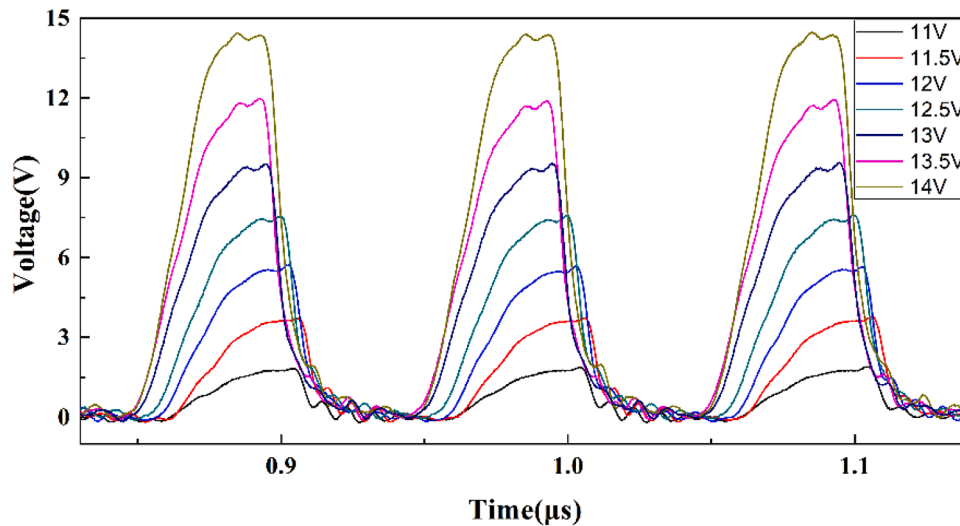


Fig. 14. The cathode voltage waveform of the laser diode load at repetition frequency of 10 MHz.

the following tests, 8 switches in parallel were employed.

3.4. Optical performance test of LD devices driven by the circuit

The characteristics of the laser output of the laser diode array was measured. A photo detector (PDA05CF2, Thorlabs) was used to measure the pulse waveforms of the laser, acquired with the oscilloscope. As the shown in Fig. 17, the laser pulses share a similar rising process, and yet some of them begin to fall because of the limited pulse widths. The rise time of the laser pulse is relatively long, due to the large stray inductance and junction capacitance. And the relatively slow process of population

inversion inside the large laser diodes in the array. As a result, the peak power of the pulses with short widths did not reach the maximum value of the large-width ones. Although the pulses fall rapidly at the end, they are not completely stopped at once. The higher the peak power of the pulse reaches, the larger the tail is. This is possibly due to a larger number of the inversed population to be recombined, and more energy stored in stray inductance needs to be discharged.

The peak power curves of the laser output at various repetition frequencies and supply voltages, with a fixed burst repetition frequency and width of 10 Hz of 100 μ s, respectively, are shown in Fig. 18. When the supply voltage is greater than 10.5 V, the laser diode starts to switch

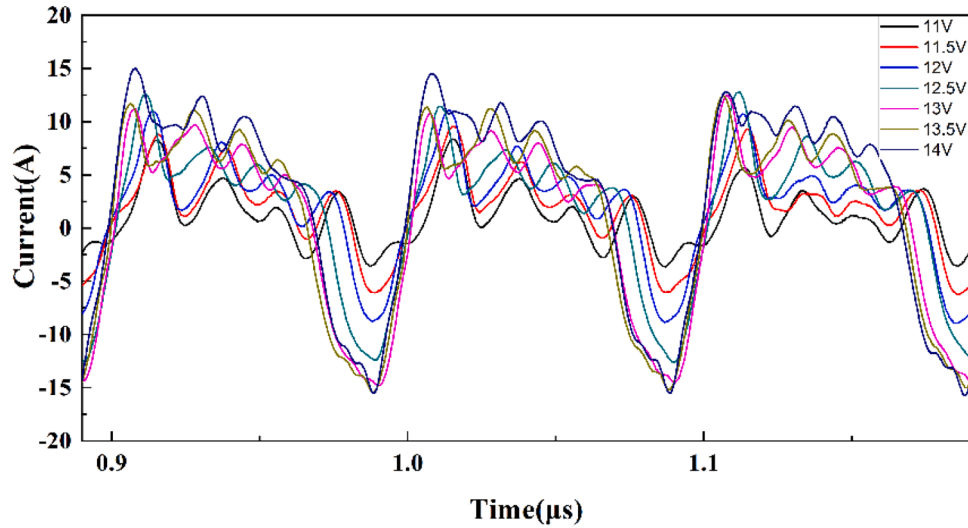


Fig. 15. The current waveform of the laser diode load at repetition frequency of 10 MHz.

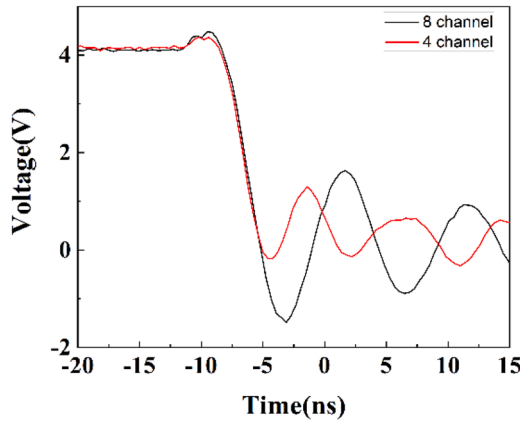


Fig. 16. Waveform at the output showing the falling time.

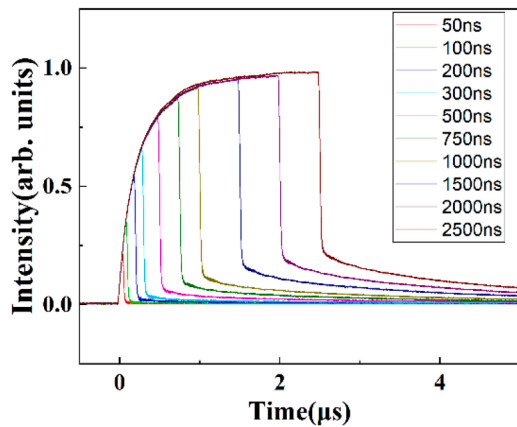


Fig. 17. Waveform diagram of laser pulse at different pulse width.

on and the output power is proportional to the supply voltage. When the voltage is lower than this voltage, the laser diode output power is almost zero. With the decrease of repetition frequency, the output peak power of light becomes stronger under the same driving voltage. When the repetition frequency is low, the pulse duration is long and establish a stable current output. Thus, the laser output power at a lower repetition

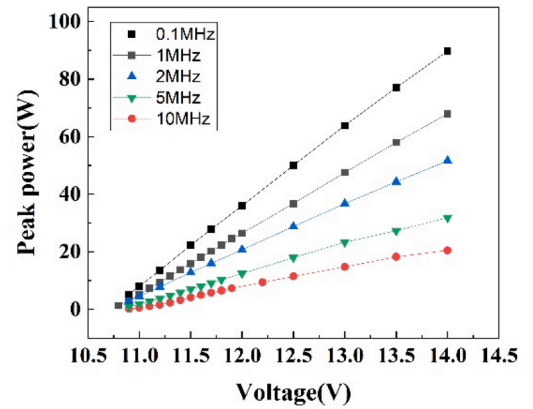


Fig. 18. Pulse peak power curve of laser diodes at different repetition frequency.

frequency is greater than that at a higher repetition frequency, though the duty cycle is fixed at 50 %. When the driver voltage is 14 V and the repetition frequency is 0.1 MHz, the maximum peak power reached the maximum value of 90 W.

In order to test the stability of the driving circuit, we measured the laser output power at repetition frequencies of 10 MHz under 13 V. As can be seen from Fig. 19, at the beginning of the test, the thermal equilibrium of the system was not yet established, and output power presented the downward trend. After the thermal stability state is established, the system exhibits extremely high stability. The output power stability was 0.49 %.

3.5. Testing of the drive circuit under higher repetition frequency conditions

A fast pulsed laser diode (PLD-100W-M) was selected as the load to verify if the driver is capable of generating high-speed pulse stably. The laser diode supports a maximum peak laser power of 100 W with a pulse width of 1 ns to 100 ns. We tested the laser at a supply voltage of 9 V, a repetition frequency of 10 MHz, and a pulse width of 30 ns. Fig. 20(a) is the voltage waveform at the cathode, Fig. 20(b) is the current waveform, and Fig. 20(c) is the laser output waveform of the laser diode. It can be seen from the results that the laser can work normally under the condition above.

In order to verify the characteristics of the pulse driver at higher

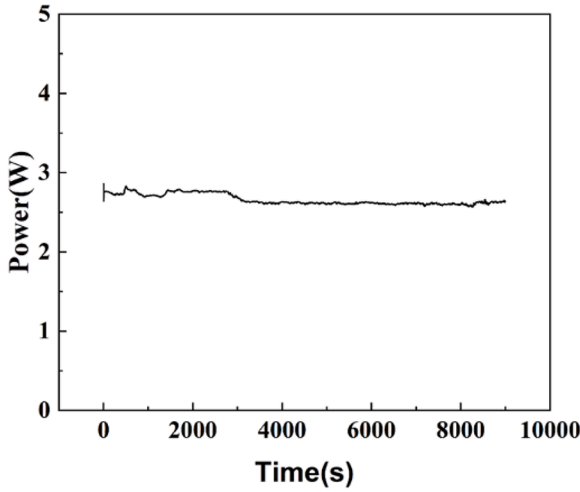


Fig. 19. Laser output power variation at 10 ns and 10 MHz across 15 min.

repetition frequencies, we tested the laser output power at different repetition frequencies at the voltage of 10 V and the pulse width of 10 ns. As shown in Fig. 21, when the driving pulse reaches 40 MHz, the circuit operates at a critical state in which the switch can barely fully turn off. As shown in Fig. 22, when the repetition frequency is <40 MHz, the output power is proportional to the repetition frequency. However, the average output power of the laser diode increases dramatically when the repetition frequency is 50 MHz. Since the reversed population of carriers in the output of the laser diode could not be completely recombined within several nanoseconds, and hence the output cannot be completely cut off within 10 ns before it is turned on in the next period at the repetition frequency 50 MHz. Therefore, the experiment proves that the pulse driver has the ability to work at 40 MHz repetition frequency.

In all, the system can operate continuously under conditions of high repetition frequency and high duty cycle. It has to be pointed out that

the laser under the high-repetition-frequency operation state generates much more heat than that in the low-repetition-frequency operation state. And, as a result, the heat-dissipation capability of the laser diode array actually limits the peak output power of the laser array. Nevertheless, since this laser driver power supply is used for lidar with high-frequency pulse coding and single-photon detectors, it is still possible to achieve high-precision lidar detection with a relatively low peak power.

Table 2 depicts a comparison of the performance of various driving technologies. Compared with previous works, the proposed pulse driver circuit in this paper has a wider range of pulse width adjustment and a higher repetition frequency. And it can achieve both relatively high peak power and average power.

4. Conclusion

In this article, a laser pulse power system is proposed. The system employs a frequency synthesis method in order to achieve a high repetition frequency. A voltage and current-adjustable DC-DC converter is used to regulate the DC power parameters of the laser power supply system. Various loads, such as a small resistor, a single laser diode, and a laser diode array, are used to measure the output characteristics of the power supply. Experimental results show that the power supply can achieve a repetition frequency of 10 MHz across different loads, with the peak pulse output current reaching 10 A at 50 % duty cycle. The pulse power system was evaluated at various repetition frequency up to 40 MHz, and has the potential to reach a higher value with a faster medium-power laser diode. Although the junction capacitance of the switching transistors increased, it was found during the test that the change in the number of transistors did not affect the rising edge time of the driving waveform. Moreover, in the optical test, it was discovered that the waveform oscillation only occurred in the electrical test and did not have an impact on the output waveform and power of the laser. Therefore, we believe that this multiple-parallel-switch circuit structure can effectively enhance the output power and repetition frequency of the driving circuit, and will not have an impact on the optical output of

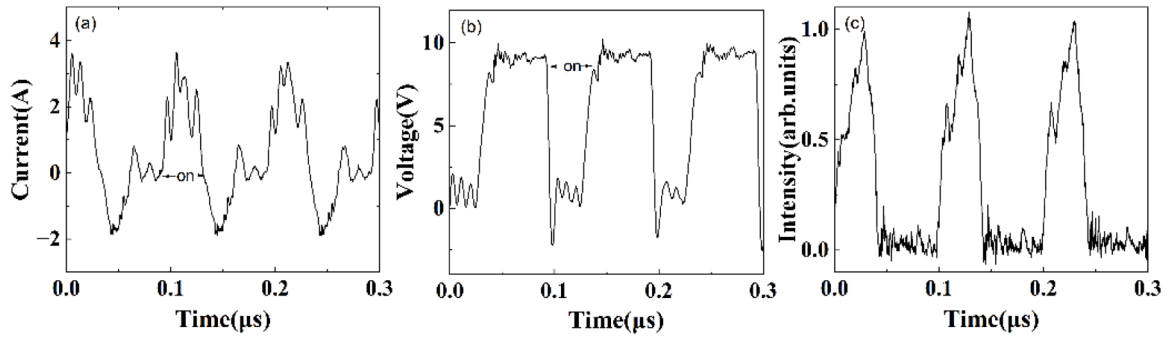


Fig. 20. Electrical characteristics of fast laser diode and optical output waveform with the driver.

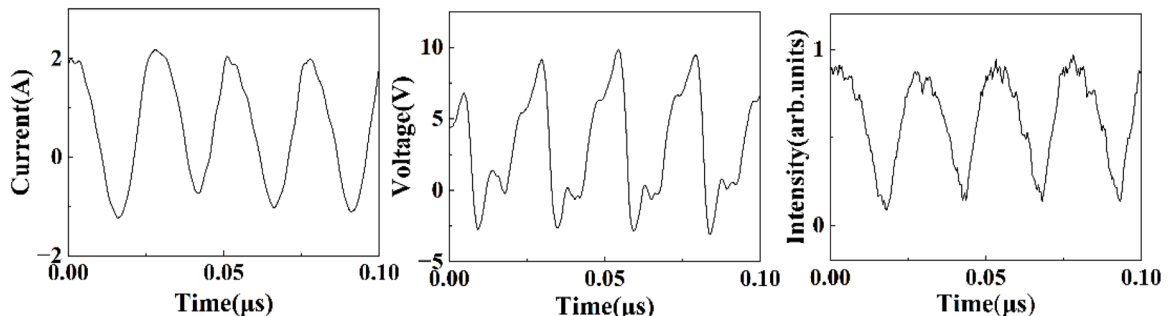


Fig. 21. Electrical characteristics of fast laser diode and optical output waveform with the driver under 40 MHz.

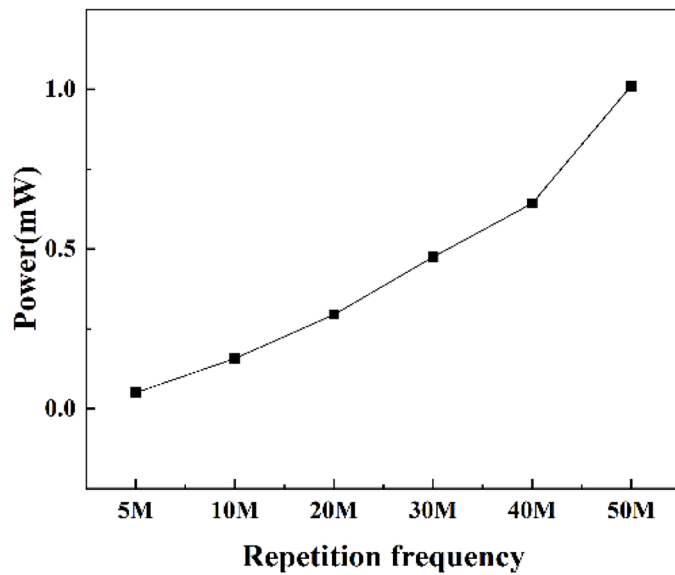


Fig. 22. Power curve of laser at different repetition frequency.

Table 2
Experimental performance comparison of pulse driver circuit technologies.

Year	Power transistor	Repetition frequency	Pulse width	Peak power
2010 [30]	MOSFET	1 kHz - 10 kHz	3 - 20 ns	–
2010 [31]	MOSFET	50 kHz	12 ns	30 W
2010 [32]	MOSFET	10 Hz - 1 MHz	16 ns-1 μ s	–
2019 [17]	MOSFET	5 kHz - 25 MHz	3.5 - 5.1 ns	36.8 mW
2018 [33]	MOSFET	500 kHz - 100 MHz	538 ps - 10.27 ns	98 mW
2021 [29]	GaN	1 kHz	6 ns	880 W
2022 [28]	GaN	10 MHz	10 - 100 ns	150 W
This work	GaN	1 Hz - 40 MHz	5 ns - 1 μ s	90 W

the laser. This system can achieve higher output power, repetition frequency and wider adjustable pulse width while using commercial GaN driver transistor, without affecting the optical output of the laser.

It is important to note that, in this design, the output current is primarily controlled by the voltage. Although the pulse current signal is monitored in this design, feedback control via current signal sampling remains challenging at high repetition frequencies. A feedback scheme based on pulsed current is currently under investigation. In addition, during the operation with a higher repetition frequency, there may be a situation where the switching transistor fails to be completely turned off, which causes the laser to malfunction. To address this drawback, GaN transistor with smaller parasitic capacitances should be chosen and reduce the parasitic inductance on the circuit board.

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Supplementary materials: See Supplementary Material for supporting content.

CRediT authorship contribution statement

Bin Wu: Writing – original draft, Validation, Methodology, Investigation, Data curation. **Junliang Liu:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Gaoyou Liu:** Supervision. **Fengxin Dong:** Investigation. **Xiangqian Li:** Investigation. **Cheng Fei:** Formal analysis. **Yongfu Li:** Formal analysis. **Zhaojun Liu:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rineng.2025.107983](https://doi.org/10.1016/j.rineng.2025.107983).

Data availability

No data was used for the research described in the article.

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